

Module 2 | Approximation Theory



So far, we have established (or have seen strong evidence of) the following results:

- Given $f \in C^0[a, b]$, we can find polynomials of sufficiently large degree that are arbitrarily close to f .
- Given $f \in C^0[a, b]$, we can find unique polynomials of degree n that interpolates f at a given set of $(n + 1)$ points.
- Given $f \in C^0[a, b]$, we can choose the interpolation points in a way so that the f -interpolant polynomials p_n behave “nicely” (e.g., converge uniformly to f). In fact, the Chebyshev polynomials work for all but the ugliest function in $C[a, b]$.

However, Interpolation is not the only way to approximate a function. In this chapter, we investigate alternative approaches that directly optimize the quality of the approximation. In particular, we want to ask

Given $f \in C^0[a, b]$, what is the “best” polynomial $p_n \in \mathcal{P}_n$ that approximates f ?

Let’s start by defining what we mean by “best”. In some sense, we are trying to minimize some kind of error function. Let’s start with a quick Linear Algebra detour.

§2.1 Normed Vector Space and Inner-Product Space

Definition 2.1.9

Given a vector space X over a subfield $\mathbb{F}$ of the complex numbers $\mathbb{C}$, a norm on X is a real-valued function $\|\cdot\| : X \rightarrow \mathbb{R}$ with the following properties,

$$\text{Triangle inequality: } \|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\| \quad \text{for every } \vec{x}, \vec{y} \in X, \quad (\text{N1})$$

$$\text{Absolute homogeneity: } \|\alpha \vec{x}\| = |\alpha| \|\vec{x}\| \quad \text{for every } \vec{x} \in \mathbb{R}^n \text{ and every } \alpha \in \mathbb{R}, \quad (\text{N2})$$

$$\text{Positive definiteness: } \|\vec{x}\| \geq 0 \quad \text{for every } \vec{x} \in X \text{ and } \|\vec{x}\| = 0 \implies \vec{x} = \vec{0}. \quad (\text{N3})$$

Example 2.1.10

Let $\vec{x} = \langle x_1, x_2, \dots, x_n \rangle \in \mathbb{R}^n$. There are three common examples of vector norm:

(a) The ℓ_2 -norm $\|\vec{x}\|_2 := \sqrt{\sum_{k=1}^n x_k^2} = \sqrt{\vec{x}^T \vec{x}}$.

(b) The ℓ_1 -norm $\|\vec{x}\|_1 := \sum_{k=1}^n |x_k|$

(c) The ℓ_∞ -norm $\|\vec{x}\|_\infty := \max_{k=1, \dots, n} |x_k|$

The norms in the example above are all special cases of the ℓ_p -norm,

$$\|\vec{x}\|_p = \left(\sum_{k=1}^n |x_k|^p \right)^{1/p}$$

which is a norm for any real number $p \geq 1$. Increasing p means that more and more emphasis is given to the maximum element $|x_k|$.

■ **Question 24.**

□

Consider the vectors $\vec{a} = (1, -2, 3)^T$, $\vec{b} = (2, 0, -1)^T$, and $\vec{c} = (0, 1, 4)^T$. Evaluate their ℓ_1 -, ℓ_2 -, and ℓ_∞ -norms.

■ **Question 25.**

□

Sketch the “unit circles” $S_p = \{\vec{x} \in \mathbb{R}^2 : \|\vec{x}\|_p = 1\}$ for $p = 1, 1.5, 2, \infty$.

■ **Question 26.**

□

Use the triangle inequality to show that for an arbitrary norm $\|\cdot\|$ on $\mathbb{R}^n$, there is a finite constant K such that $\|x\| \leq K\|x\|_\infty$ for all $x \in \mathbb{R}^n$.

Definition 2.1.11

An inner product on a vector space V is a function which assigns to each ordered pair of vectors $(\vec{u}, \vec{v})$ in V a scalar $\langle \vec{u}, \vec{v} \rangle \in \mathbb{F}$ such that

- $\langle \vec{u} + \alpha \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \alpha \langle \vec{v}, \vec{w} \rangle$ for $\alpha \in \mathbb{F}$, (bi-linearity)
- $\langle \vec{u}, \vec{v} \rangle > 0$ for $\vec{v} \neq \vec{0}$
- $\langle \vec{v}, \vec{u} \rangle = \overline{\langle \vec{u}, \vec{v} \rangle}$, where the bar denotes complex conjugate

Example 2.1.12

The standard inner product on $\mathbb{F}^n$ is defined as

$$\langle \vec{u}, \vec{v} \rangle = \vec{v}^* \vec{u} = \sum_{i=1}^n u_i \overline{v_i}$$

■ **Question 27.**

□

Suppose $\vec{u}, \vec{v} \in \mathbb{R}^2$. Show that

$$\langle \vec{u}, \vec{v} \rangle = u_1 v_1 - u_2 v_1 - u_1 v_2 + 4u_2 v_2$$

defines an inner product on $\mathbb{R}^2$.

Every inner-product space is a normed vector space with the induced norm defined as

$$\|\vec{v}\| = \sqrt{\langle \vec{v}, \vec{v} \rangle}$$